

EmbedUR Networks Training

Assessment – 3

Module – 5: Layer 3 – IP Address and DHCP Protocol

&

Module – 6: Routing and Packet Flow

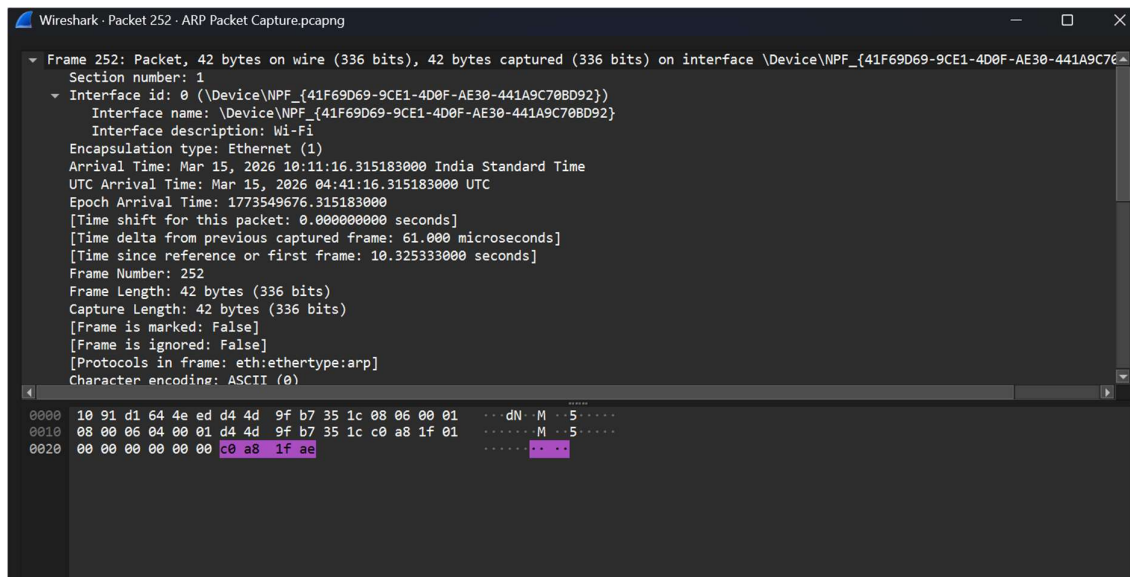
Name: Kishore S

Module - 5

1) Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames, and discuss the role of the sender's IP and MAC address in these packets.

ARP Packets Capture:

```
▼ Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: SkyworthDigi_b7:35:1c (d4:4d:9f:b7:35:1c)
  Sender IP address: 192.168.31.1
  Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
  Target IP address: 192.168.31.174
```



```
▼ Ethernet II, Src: SkyworthDigi_b7:35:1c (d4:4d:9f:b7:35:1c), Dst: Intel_64:4e:ed (10:91:d1:64:4e:ed)
  Destination: Intel_64:4e:ed (10:91:d1:64:4e:ed)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Source: SkyworthDigi_b7:35:1c (d4:4d:9f:b7:35:1c)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 0]
```

2) Using Packet Tracer, simulate an ARP spoofing attack. Analyze the behavior of devices on the network when they receive a malicious ARP response.

ARP Spoofing:

Initial Router Config:

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
*Mar 18 10:25:18.975: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
R1#
R1#
*Mar 18 10:25:19.959: %SYS-5-CONFIG_I: Configured from console by console
*Mar 18 10:25:19.975: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1#
R1#sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.168.1.1     YES manual up          up
FastEthernet1/0          unassigned      YES unset  administratively down down
FastEthernet2/0          unassigned      YES unset  administratively down down
FastEthernet2/1          unassigned      YES unset  administratively down down
Serial3/0                unassigned      YES unset  administratively down down
Serial3/1                unassigned      YES unset  administratively down down
Serial3/2                unassigned      YES unset  administratively down down
Serial3/3                unassigned      YES unset  administratively down down
Serial3/4                unassigned      YES unset  administratively down down
Serial3/5                unassigned      YES unset  administratively down down
Serial3/6                unassigned      YES unset  administratively down down
Serial3/7                unassigned      YES unset  administratively down down
Ethernet4/0              unassigned      YES unset  administratively down down
Ethernet4/1              unassigned      YES unset  administratively down down
Ethernet4/2              unassigned      YES unset  administratively down down
Ethernet4/3              unassigned      YES unset  administratively down down
Ethernet4/4              unassigned      YES unset  administratively down down
Ethernet4/5              unassigned      YES unset  administratively down down
--More--
```

Victim (Ubuntu Linux) PC Config:

```
QEMU (Victim) - TightVNC Viewer
Terminal
osboxes@osboxes: ~
Search your computer
osboxes@osboxes:~$ sudo ip addr add 192.168.1.20/24 dev ens3
osboxes@osboxes:~$ sudo ip link set ens3 up
osboxes@osboxes:~$ sudo ip route add default via 192.168.1.1
osboxes@osboxes:~$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
    group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP mo
    de DEFAULT group default qlen 1000
    link/ether 0c:e2:57:02:00:00 brd ff:ff:ff:ff:ff:ff
osboxes@osboxes:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP gr
    oup default qlen 1000
    link/ether 0c:e2:57:02:00:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.20/24 scope global ens3
        valid_lft forever preferred_lft forever
    inet6 fe80::e2:57ff:fe02:0/64 scope link
```

Attacker (Kali Linux) PC Config:

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali㉿kali)-[~]  
$ sudo ip addr add 192.168.1.20/24 dev eth0  
  
(kali㉿kali)-[~]  
$ sudo ip link set eth0 up  
  
(kali㉿kali)-[~]  
$ sudo ip route add default via 192.168.1.1  
  
(kali㉿kali)-[~]  
$ ip route  
default via 192.168.1.1 dev eth0  
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.20  
  
(kali㉿kali)-[~]  
$ ping 192.168.1.1  
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data:  
64 bytes from 192.168.1.1: icmp_seq=1 ttl=255 time=84.9 ms  
64 bytes from 192.168.1.1: icmp_seq=2 ttl=255 time=14.6 ms  
64 bytes from 192.168.1.1: icmp_seq=3 ttl=255 time=34.7 ms  
64 bytes from 192.168.1.1: icmp_seq=4 ttl=255 time=48.6 ms  
64 bytes from 192.168.1.1: icmp_seq=5 ttl=255 time=26.4 ms  
^C  
— 192.168.1.1 ping statistics —  
5 packets transmitted, 5 received, 0% packet loss, time 4044ms  
rtt min/avg/max/mdev = 14.575/41.832/84.874/24.206 ms
```

ARP Spoofing from Kali:

The screenshot displays two side-by-side terminal windows from a Kali Linux desktop environment. The left window shows the execution of `sudo sysctl -w net.ipv4.ip\_forward=1` followed by `sudo arpspoof -i eth0 -t 192.168.1.10 192.168.1.1`, which triggers a series of ARP reply messages from the host to the target IP. The right window shows the same `arpspoof` command being run again, continuing the stream of ARP replies. Both terminals show standard Linux prompts and system status indicators at the top.


```

R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#int f1/0
R1(config-if)#ip route 192.168.2.0 255.255.255.2 10.0.0.2
%Inconsistent address and mask
R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)#
R1(config)#exit
R1#exit
*Mar 17 20:18:37.059: %SYS-5-CONFIG_I: Configured from console by console
R1#exit

```

R2:

```

R2#en
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#exit
R2(config)#exit
R2#
R2#
*Mar 17 20:21:28.687: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
R2#en
*Mar 17 20:21:29.671: %SYS-5-CONFIG_I: Configured from console by console
*Mar 17 20:21:29.687: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2#en
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#ip address 10.0.0.2 255.255.255.252
^
% Invalid input detected at '^' marker.

R2(config)#int f1/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#exit
R2(config)#exit
R2#
*Mar 17 20:22:36.043: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 17 20:22:37.043: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R2(config)#exit
R2#
R2#sh i
*Mar 17 20:22:39.899: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int brief

```

```

R2#sh ip int brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.2.1	YES	manual	up	up
FastEthernet1/0	10.0.0.2	YES	manual	up	up
FastEthernet2/0	unassigned	YES	unset	administratively down	down
FastEthernet2/1	unassigned	YES	unset	administratively down	down
Serial3/0	unassigned	YES	unset	administratively down	down
Serial3/1	unassigned	YES	unset	administratively down	down
Serial3/2	unassigned	YES	unset	administratively down	down
Serial3/3	unassigned	YES	unset	administratively down	down
Serial3/4	unassigned	YES	unset	administratively down	down
Serial3/5	unassigned	YES	unset	administratively down	down
Serial3/6	unassigned	YES	unset	administratively down	down
Serial3/7	unassigned	YES	unset	administratively down	down
Ethernet4/0	unassigned	YES	unset	administratively down	down
Ethernet4/1	unassigned	YES	unset	administratively down	down
Ethernet4/2	unassigned	YES	unset	administratively down	down
Ethernet4/3	unassigned	YES	unset	administratively down	down
Ethernet4/4	unassigned	YES	unset	administratively down	down
Ethernet4/5	unassigned	YES	unset	administratively down	down
Ethernet4/6	unassigned	YES	unset	administratively down	down
Ethernet4/7	unassigned	YES	unset	administratively down	down
Serial5/0	unassigned	YES	unset	administratively down	down
Serial5/1	unassigned	YES	unset	administratively down	down
Serial5/2	unassigned	YES	unset	administratively down	down
Serial5/3	unassigned	YES	unset	administratively down	down
POS6/0	unassigned	YES	unset	administratively down	down

```

R2#
R2#
R2#
R2#en
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f1/0
R2(config-if)#
R2(config-if)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
R2(config)#

```

PC Configs:

```

Welcome to Virtual PC Simulator, version 0.0.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpc.ssf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> show ip

NAME       : PC1[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20010
RHOST:PORT  : 127.0.0.1:20011
MTU         : 1500

```

```

ase visit wiki.freecode.c
om.cn.

Press '?' to get help.

Executing the startup fil
e

PC2> ip 192.168.1.20/24 192.168.1.1
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0 gateway 192.168.1.1

PC2> show ip

NAME       : PC2[1]
IP/MASK     : 192.168.1.20/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 20012
RHOST:PORT  : 127.0.0.1:20013
MTU         : 1500

```

```

om.cn.

Press '?' to get help.

Executing the startup fil
e

PC3>
PC3>
PC3>
PC3> ip 192.168.2.10/24 192.168.2.1
Checking for duplicate address...
PC3 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1

PC3> show ip

NAME       : PC3[1]
IP/MASK     : 192.168.2.10/24
GATEWAY     : 192.168.2.1
DNS         :
MAC         : 00:50:79:66:68:02
LPORT      : 20014
RHOST:PORT  : 127.0.0.1:20015
MTU         : 1500

```

```

Press '?' to get help.

Executing the startup file

PC4> ip 192.168.2.10/24 192.168.2.1
Checking for duplicate address...
192.168.2.10 is being used by MAC 00:50:79:66:68:02
Address not changed

PC4> ip 192.168.2.20/24 192.168.2.1
Checking for duplicate address...
PC4 : 192.168.2.20 255.255.255.0 gateway 192.168.2.1

PC4> show ip

NAME       : PC4[1]
IP/MASK     : 192.168.2.20/24
GATEWAY     : 192.168.2.1
DNS         :
MAC         : 00:50:79:66:68:03
LPORT      : 20016
RHOST:PORT  : 127.0.0.1:20017
MTU         : 1500

```

Ping Test:

Ping from PC1 from Network 1 to PCs from Other Network:

```

MTU       : 1500

PC1> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=366.250 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=258.286 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=355.425 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=219.034 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=336.805 ms

PC1> ping 192.168.2.20

84 bytes from 192.168.2.20 icmp_seq=1 ttl=62 time=467.764 ms
84 bytes from 192.168.2.20 icmp_seq=2 ttl=62 time=267.046 ms
84 bytes from 192.168.2.20 icmp_seq=3 ttl=62 time=273.848 ms
84 bytes from 192.168.2.20 icmp_seq=4 ttl=62 time=187.112 ms
84 bytes from 192.168.2.20 icmp_seq=5 ttl=62 time=299.350 ms

PC1> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=254 time=275.857 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=254 time=151.767 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=254 time=213.930 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=254 time=238.574 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=254 time=156.326 ms

PC1>

```

Ping from PC3 from Network 2 to Other Network PCs:

```
PC3> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=290.596 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=185.933 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=266.102 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=348.677 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=195.259 ms

PC3> ping 192.168.1.20

84 bytes from 192.168.1.20 icmp_seq=1 ttl=62 time=279.252 ms
84 bytes from 192.168.1.20 icmp_seq=2 ttl=62 time=188.127 ms
84 bytes from 192.168.1.20 icmp_seq=3 ttl=62 time=222.349 ms
84 bytes from 192.168.1.20 icmp_seq=4 ttl=62 time=207.042 ms
84 bytes from 192.168.1.20 icmp_seq=5 ttl=62 time=305.665 ms

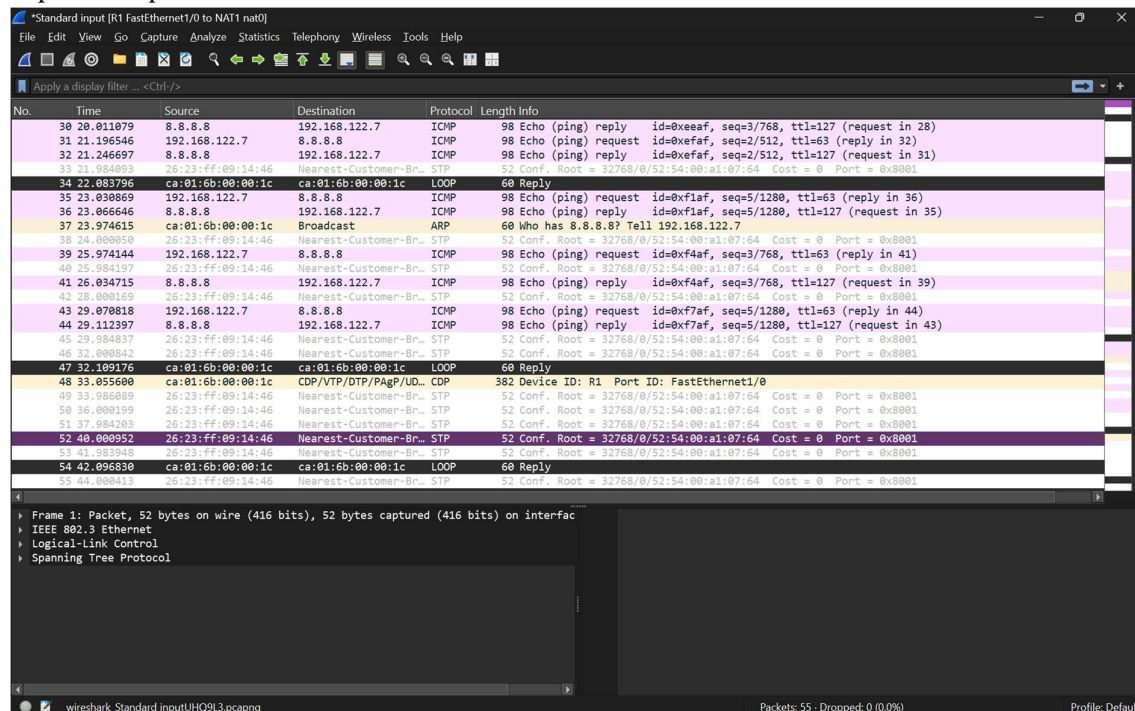
PC3> ping 1992.168.1.1
Cannot resolve 1992.168.1.1

PC3> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=254 time=177.035 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=254 time=201.588 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=254 time=319.535 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=254 time=180.419 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=254 time=200.464 ms

PC3> 
```

4) Use Wireshark to capture DHCP Discover, Offer, Request, and Acknowledge messages and explain the process.



Assign IP addresses from these subnets to devices in Cisco Packet Tracer and verify connectivity using ping between them.

192.168.1.0/24

Subnet Mask: 255.255.255.0

192.168.1.0 0 0 0 0 0 0 0 0
192.168.1.0 0 1 0 0 0 0 0 0
192.168.1.0 1 0 0 0 0 0 0 0
192.168.1.0 1 1 0 0 0 0 0 0

8b 8b 8b 2b = 26bit

Subnet-1: 192.168.1.00 0 0 0 0 0 0 0 0
192.168.1.0 to (00111111)₂
192.168.1.63 (63)₁₀

Network ID: 192.168.1.0/26
Broadcast ID: 192.168.1.63/26
(01000000)₂ = (64)₁₀

Subnet-2: 192.168.1.01 0 0 0 0 0 0 0 0
192.168.1.64 to (01111111)₂
192.168.1.127 = (127)₁₀

Network ID: 192.168.1.64/26
Broadcast: 192.168.1.127/26

Subnet-3: 192.168.1.10 0 0 0 0 0 0 0 0
192.168.1.128 to (10000000)₂ = (128)₁₀
192.168.1.191 (10111111)₂ = (191)₁₀

Network ID: 192.168.1.128/26
Broadcast: 192.168.1.191/26

Subnet 4: 192.168.1. 11 00 00 00 0 0

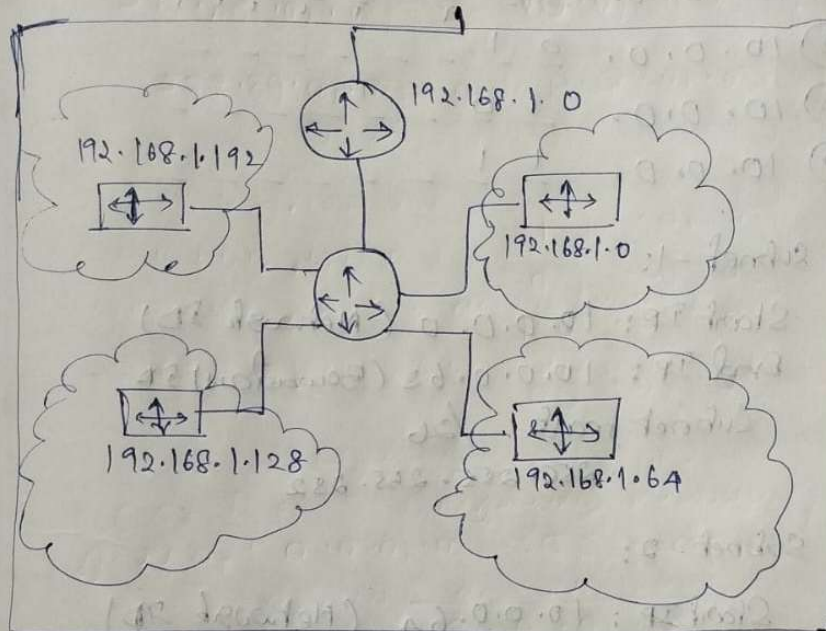
$$(11000000)_2 = (192)_{10}$$

$$(11111111)_2 = (252)_{10}$$

$\therefore$ 192.168.1. 192 to
192.168.1. 252

Network ID : 192.168.1. 192

Broadcast : 192.168.1. 252

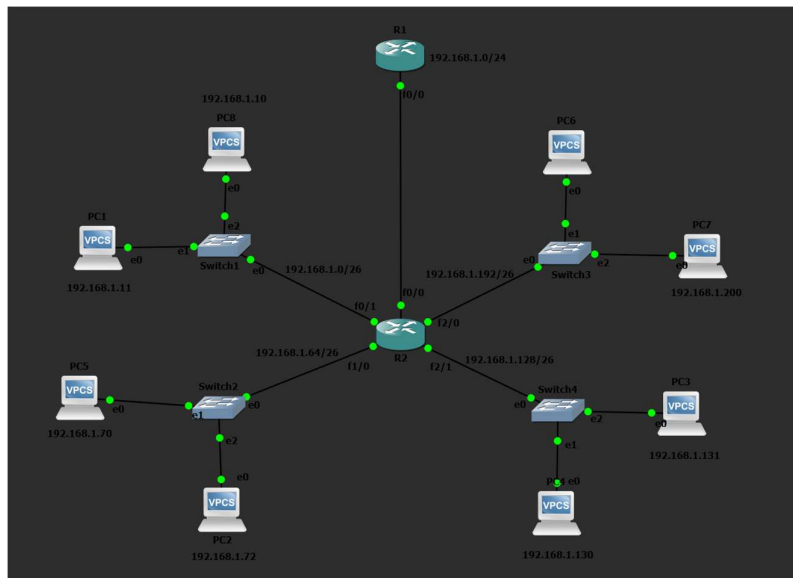


Subnet mask: 255.255.255.11000000

$$\begin{array}{r} 255.255.255.11000000 \\ \hline 192 \end{array}$$

255.255.255.192

Topology:



Configurations:

R1(External Router) - Config:

```
R1
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#interface f0/0
R1(config-if)#
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
R1#
*Mar 16 06:41:57.643: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int brief
Interface          IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0    10.0.0.1        YES manual  up          up
FastEthernet0/1    unassigned      YES unset   administratively down down
R1#
```

R2 - Main Network to External Link Config:

```
R2#
R2#en
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#interface f0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#exi
R2(config)#exit
R2#
R2#sh
*Mar 16 06:45:08.291: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          10.0.0.2        YES manual up          up
FastEthernet0/1          192.168.1.1     YES manual administratively down down
FastEthernet1/0          unassigned      YES unset  administratively down down
FastEthernet2/0          unassigned      YES unset  administratively down down
FastEthernet2/1          unassigned      YES unset  administratively down down
FastEthernet3/0          unassigned      YES unset  administratively down down
Serial4/0                unassigned      YES unset  administratively down down
Serial4/1                unassigned      YES unset  administratively down down
Serial4/2                unassigned      YES unset  administratively down down
Serial4/3                unassigned      YES unset  administratively down down
Serial4/4                unassigned      YES unset  administratively down down
Serial4/5                unassigned      YES unset  administratively down down
Serial4/6                unassigned      YES unset  administratively down down
Serial4/7                unassigned      YES unset  administratively down down
GigabitEthernet5/0       unassigned      YES unset  administratively down down
```

R2 - Subnet - 1 Config:

```
R2#
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#interface f0/1
R2(config-if)#
R2(config-if)#ip address 192.168.1.1 255.255.255.192
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#ext
*Mar 16 06:46:45.251: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 16 06:46:46.251: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R2(config-if)#exit
R2(config)#
R2(config)#exit
R2#
R2#
R2#
*Mar 16 06:46:52.527: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          10.0.0.2        YES manual up          up
FastEthernet0/1          192.168.1.1     YES manual up          up
FastEthernet1/0          unassigned      YES unset  administratively down down
FastEthernet2/0          unassigned      YES unset  administratively down down
FastEthernet2/1          unassigned      YES unset  administratively down down
FastEthernet3/0          unassigned      YES unset  administratively down down
Serial4/0                unassigned      YES unset  administratively down down
Serial4/1                unassigned      YES unset  administratively down down
Serial4/2                unassigned      YES unset  administratively down down
```

R2 - Subnet - 2 Config:

```
R2#
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#interface f1/0
R2(config-if)#ip address 192.168.1.65 255.255.255.192
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#
*Mar 16 06:48:33.907: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 16 06:48:34.907: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R2(config-if)#exit
R2(config)#exit
R2#
R2#
*Mar 16 06:48:37.751: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          10.0.0.2        YES manual up          up
FastEthernet0/1          192.168.1.1     YES manual up          up
FastEthernet1/0          192.168.1.65    YES manual up          up
FastEthernet2/0          unassigned      YES unset  administratively down down
FastEthernet2/1          unassigned      YES unset  administratively down down
FastEthernet3/0          unassigned      YES unset  administratively down down
Serial4/0                unassigned      YES unset  administratively down down
Serial4/1                unassigned      YES unset  administratively down down
Serial4/2                unassigned      YES unset  administratively down down
Serial4/3                unassigned      YES unset  administratively down down
Serial4/4                unassigned      YES unset  administratively down down
--More--
```

R2 - Subnet - 3 Config:

```
R2
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#interface f2/1
R2(config-if)#ip address 192.168.1.129 255.255.255.192
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#exit
R2(config)#exit
R2#
*Mar 16 06:50:18.375: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
R2#
*Mar 16 06:50:19.375: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
*Mar 16 06:50:19.623: %SYS-5-CONFIG_I: Configured from console by console
R2#
R2#sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          10.0.0.2        YES manual up          up
FastEthernet0/1          192.168.1.1     YES manual up          up
FastEthernet1/0          192.168.1.65    YES manual up          up
FastEthernet2/0          unassigned      YES unset  administratively down down
FastEthernet2/1          192.168.1.129   YES manual up          up
FastEthernet3/0          unassigned      YES unset  administratively down down
Serial4/0                unassigned      YES unset  administratively down down
Serial4/1                unassigned      YES unset  administratively down down
Serial4/2                unassigned      YES unset  administratively down down
Serial4/3                unassigned      YES unset  administratively down down
Serial4/4                unassigned      YES unset  administratively down down
--More--
```

R2 - Subnet - 4 Config:

```
R2
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#interface f2/0
R2(config-if)#ip address 192.168.1.193 255.255.255.192
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#exit
R2(config)#exit
R2#
*Mar 16 06:52:24.203: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Mar 16 06:52:25.203: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R2(config)#exit
R2#
*Mar 16 06:52:25.991: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          10.0.0.2        YES manual up          up
FastEthernet0/1          192.168.1.1     YES manual up          up
FastEthernet1/0          192.168.1.65    YES manual up          up
FastEthernet2/0          192.168.1.193   YES manual up          up
FastEthernet2/1          192.168.1.129   YES manual up          up
FastEthernet3/0          unassigned      YES unset  administratively down down
Serial4/0                unassigned      YES unset  administratively down down
Serial4/1                unassigned      YES unset  administratively down down
Serial4/2                unassigned      YES unset  administratively down down
Serial4/3                unassigned      YES unset  administratively down down
Serial4/4                unassigned      YES unset  administratively down down
--More--
```

PC Configuration Examples:

```
Kishore_PC - PuTTY

For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.168.1.10/26 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.192 gateway 192.168.1.1

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.1.10/26
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10028
RHOST:PORT : 127.0.0.1:10029
MTU        : 1500

PC1>
```

```
Kishore_PC - PuTTY

Executing the startup file

PC1> ip 192.168.1.70/26 192.168.1.64
Invalid gateway address

PC1> ip 192.168.1.70/26 192.168.1.65
Checking for duplicate address...
PC1 : 192.168.1.70 255.255.255.192 gateway 192.168.1.65

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.1.70/26
GATEWAY    : 192.168.1.65
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10030
RHOST:PORT : 127.0.0.1:10031
MTU        : 1500

PC1>
```

```
192.168.11.128 - PuTTY

For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC5> ip 192.168.1.70/26 192.168.1.65
Checking for duplicate address...
PC5 : 192.168.1.70 255.255.255.192 gateway 192.168.1.65

PC5> show ip

NAME       : PC5[1]
IP/MASK    : 192.168.1.70/26
GATEWAY    : 192.168.1.65
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 20010
RHOST:PORT : 127.0.0.1:20011
MTU        : 1500

PC5>
```

```
Kishore_PC - PuTTY

For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2> ip 192.168.1.72/26 192.168.1.65
Checking for duplicate address...
PC1 : 192.168.1.72 255.255.255.192 gateway 192.168.1.65

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 192.168.1.72/26
GATEWAY    : 192.168.1.65
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10028
RHOST:PORT : 127.0.0.1:10029
MTU        : 1500

PC2>
```

```
Kishore_PC - PuTTY

Executing the startup file

PC4> ip addr 192.168.1.130/26 192.168.1.129
Invalid address

PC4> ip 192.168.1.130/26 192.168.1.129
Checking for duplicate address...
PC1 : 192.168.1.130 255.255.255.192 gateway 192.168.1.129

PC4> show ip

NAME       : PC4[1]
IP/MASK    : 192.168.1.130/26
GATEWAY    : 192.168.1.129
DNS        :
MAC        : 00:50:79:66:68:03
LPORT     : 10034
RHOST:PORT : 127.0.0.1:10035
MTU        : 1500

PC4>
```

```
Kishore_PC - PuTTY

PC3> ip 192.168.1.131/26 192.168.1.129
Checking for duplicate address...
PC1 : 192.168.1.131 255.255.255.192 gateway 192.168.1.129

PC3> show ip

NAME       : PC3[1]
IP/MASK    : 192.168.1.131/26
GATEWAY    : 192.168.1.129
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 10032
RHOST:PORT : 127.0.0.1:10033
MTU        : 1500

PC3>
PC3>
PC3>
PC3>
PC3>
PC3>
```

Ping Test:

The screenshot displays five PuTTY terminal windows, each showing the output of a ping command. The windows are titled '192.168.11.128 - PuTTY', 'Kishore_PC - PuTTY', '192.168.11.128 - PuTTY', 'Kishore_PC - PuTTY', and 'Kishore_PC - PuTTY'. Each window shows the MTU (1500) and the RHOST:PORT (127.0.0.1:10031, 127.0.0.1:10029, or 127.0.0.1:10035). The ping results show the number of bytes received, the sequence number, and the time taken for each packet to reach the destination and return. For example, the first window shows a ping to 192.168.1.131 with results like '84 bytes from 192.168.1.131 icmp_seq=1 ttl=63 time=64.316 ms'.

6) You are given three IP addresses: 10.1.1.1, 172.16.5.10, and 192.168.1.5. Task: Identify the class of each IP address (Class A, B, or C). What is the default subnet

mask for each class?

Provide the range of IP addresses for each class.

i) 10.1.1.1 – Class A

Start IP: 10.0.0.0 (Network ID)

End IP: 10.255.255.255 (Broadcast Address)

Usable Host Addresses: 10.0.0.1 to 10.255.255.254

Subnet Mask: 255.0.0.0

ii) 172.16.5.10 – Class B

Start IP: 172.16.0.0 (Network ID)

End IP: 172.16.255.255 (Broadcast Address)

Usable Host Addresses: 172.16.0.1 to 172.16.255.254

Subnet Mask: 255.255.0.0

iii) 192.168.1.5 – Class C

Start IP: 192.168.1.0 (Network ID)

End IP: 192.168.1.255 (Broadcast Address)

Usable Host Addresses: 192.168.0.1 to 192.168.1.254

Subnet Mask: 255.255.255.0

7) In Cisco Packet Tracer, create a small network with multiple devices (e.g., 2 PCs and a router). Use private IP addresses (e.g., 192.168.1.x) on the PCs and configure the router to perform NAT to allow the PCs to access the internet.

Task: Test the NAT configuration by pinging an external IP address from the PCs and capture the traffic using Wireshark.

What is the source IP address before and after NAT?

Router Config:

```
R1#
R1#
R1#
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
*Mar 18 19:01:10.351: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 18 19:01:11.351: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R1(config-if)#exit
R1(config)#exi
R1#
*Mar 18 19:01:27.539: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
*Mar 18 19:02:07.495: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar 18 19:02:08.495: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R1#
R1#
*Mar 18 19:02:08.803: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          192.168.1.1     YES manual up          up
FastEthernet0/1          192.168.2.1     YES manual up          up
FastEthernet1/0          unassigned      YES unset  administratively down down
R1#
```

Configure the NAT Interface of the Router with DHCP Protocol:

```
R1
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f1/0
R1(config-if)#
R1(config-if)#ip address dhcp
R1(config-if)#no shut
R1(config-if)#
*Mar 18 19:10:45.283: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to u
p
*Mar 18 19:10:46.283: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1
/0, changed state to up
R1(config-if)#
R1(config-if)#
*Mar 18 19:10:57.259: %DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet1/0 assigned DH
CP address 192.168.122.7, mask 255.255.255.0, hostname R1

R1(config-if)#exit
R1(config)#exit
R1#
R1#sh
*Mar 18 19:11:07.331: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int brief
Interface                IP-Address      OK? Method Status  Protoco
l
FastEthernet0/0          192.168.1.1     YES manual up      up
FastEthernet0/1          192.168.2.1     YES manual up      up
FastEthernet1/0          192.168.122.7   YES DHCP  up      up
```

PC Config:

```
192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> show ip
NAME       : PC1[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 20017
RHOST:PORT : 127.0.0.1:20018
MTU        : 1500

PC1>

192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2> ip 192.168.1.20/24 192.168.1.1
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0 gateway 192.168.1.1

PC2> show ip
NAME       : PC2[1]
IP/MASK    : 192.168.1.20/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 20011
RHOST:PORT : 127.0.0.1:20012
MTU        : 1500

PC2>

192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC3> ip 192.168.2.10/24 192.168.2.1
Checking for duplicate address...
PC3 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1

PC3> show ip
NAME       : PC3[1]
IP/MASK    : 192.168.2.10/24
GATEWAY    : 192.168.2.1
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 20013
RHOST:PORT : 127.0.0.1:20014
MTU        : 1500

PC3>

192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC4> ip 192.168.2.20/24 192.168.2.1
Checking for duplicate address...
PC4 : 192.168.2.20 255.255.255.0 gateway 192.168.2.1

PC4> show ip
NAME       : PC4[1]
IP/MASK    : 192.168.2.20/24
GATEWAY    : 192.168.2.1
DNS        :
MAC        : 00:50:79:66:68:03
LPORT     : 20015
RHOST:PORT : 127.0.0.1:20016
MTU        : 1500

PC4>
```

Configure NAT Inside Local Networks:

```
R1#
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#
R1(config-if)#ip nat inside
R1(config-if)#
*Mar 18 19:12:53.031: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, changed state to up
R1(config-if)#exit
R1(config)#
R1(config)#exit
R1#
R1#en
*Mar 18 19:13:01.675: %SYS-5-CONFIG_I: Configured from console by console
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/1
R1(config-if)#ip nat inside
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
R1#
*Mar 18 19:13:20.703: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
```

Configure Outside NAT:

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#int f1/0
R1(config-if)#ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#
R1#
*Mar 18 19:14:24.627: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Create Access List:

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)#access-list 1 permit 192.168.2.0 0.0.0.255
R1(config)#
```

Perform NAT Overload:

```
R1(config)#ip nat inside source list 1 interface FastEthernet1/0 overload
R1(config)#ip route 0.0.0.0 0.0.0.0 FastEthernet1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R1(config)#exir
```

Add Default Gateway:

```
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.122.1
R1(config)#
R1(config)#exit
R1#
R1#
*Mar 18 19:25:59.975: %SYS-5-CONFIG_I: Configured from console by console
R1#ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/91/224 ms
R1#
```

Ping Test to a Global IP from All PCs:

The image displays four terminal windows, each representing a different PC (PC1, PC2, PC3, and PC4) connected to a network at 192.168.11.128. Each window shows the results of a ping test to the global IP address 8.8.8.8. The windows are arranged in a 2x2 grid.

PC1 Terminal: Shows five successful ping attempts to 8.8.8.8. The first three attempts are successful with varying round-trip times (e.g., 78.272 ms, 58.453 ms, 124.693 ms). The last two attempts are successful with round-trip times of 53.502 ms and 61.976 ms.

PC2 Terminal: Shows five successful ping attempts to 8.8.8.8. The first three attempts are successful with varying round-trip times (e.g., 98.367 ms, 83.560 ms, 86.105 ms). The last two attempts are successful with round-trip times of 72.362 ms and 75.899 ms.

PC3 Terminal: Shows five successful ping attempts to 8.8.8.8. The first three attempts are successful with varying round-trip times (e.g., 220.165 ms, 76.976 ms, 76.976 ms). The last two attempts are successful with round-trip times of 76.976 ms and 76.976 ms.

PC4 Terminal: Shows five successful ping attempts to 8.8.8.8. The first three attempts are successful with varying round-trip times (e.g., 54.128 ms, 68.019 ms, 68.019 ms). The last two attempts are successful with round-trip times of 68.019 ms and 68.019 ms.

Packet Capture From Wireshark:

*Standard Input [R1 FastEthernet1/0 to NAT1 nat0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
30	20.011079	8.8.8.8	192.168.122.7	ICMP	98	Echo (ping) reply id=0xeeaf, seq=3/768, ttl=127 (request in 28)
31	21.196546	192.168.122.7	8.8.8.8	ICMP	98	Echo (ping) request id=0xefaf, seq=2/512, ttl=63 (reply in 32)
32	21.246697	8.8.8.8	192.168.122.7	ICMP	98	Echo (ping) reply id=0xefaf, seq=2/512, ttl=127 (request in 31)
33	21.988093	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
34	22.083796	ca:01:6b:00:00:1c	ca:01:6b:00:00:1c	LOOP	60	Reply
35	23.030869	192.168.122.7	8.8.8.8	ICMP	98	Echo (ping) request id=0xf1af, seq=5/1280, ttl=63 (reply in 36)
36	23.066646	8.8.8.8	192.168.122.7	ICMP	98	Echo (ping) reply id=0xf1af, seq=5/1280, ttl=127 (request in 35)
37	23.974615	ca:01:6b:00:00:1c	Broadcast	ARP	60	Who has 8.8.8.8? Tell 192.168.122.7
38	24.000050	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
39	25.974144	192.168.122.7	8.8.8.8	ICMP	98	Echo (ping) request id=0xf4af, seq=3/768, ttl=63 (reply in 41)
40	25.988197	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
41	26.034715	8.8.8.8	192.168.122.7	ICMP	98	Echo (ping) reply id=0xf4af, seq=3/768, ttl=127 (request in 39)
42	28.090169	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
43	29.070818	192.168.122.7	8.8.8.8	ICMP	98	Echo (ping) request id=0xf7af, seq=5/1280, ttl=63 (reply in 44)
44	29.112397	8.8.8.8	192.168.122.7	ICMP	98	Echo (ping) reply id=0xf7af, seq=5/1280, ttl=127 (request in 43)
45	29.984837	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
46	32.000842	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
47	32.109176	ca:01:6b:00:00:1c	ca:01:6b:00:00:1c	LOOP	60	Reply
48	33.055600	ca:01:6b:00:00:1c	CDP/VTP/DTP/PagP/UD...	CDP	382	Device ID: R1 Port ID: FastEthernet1/0
49	33.988089	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
50	36.000199	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
51	37.084203	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
52	40.000952	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
53	41.983948	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
54	42.096830	ca:01:6b:00:00:1c	ca:01:6b:00:00:1c	LOOP	60	Reply
55	44.000413	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001

Frame 1: Packet, 52 bytes on wire (416 bits), 52 bytes captured (416 bits) on interface

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

wireshark_Standard InputUHQ9L3.pcapng

Packets: 55 · Dropped: 0 (0.0%)

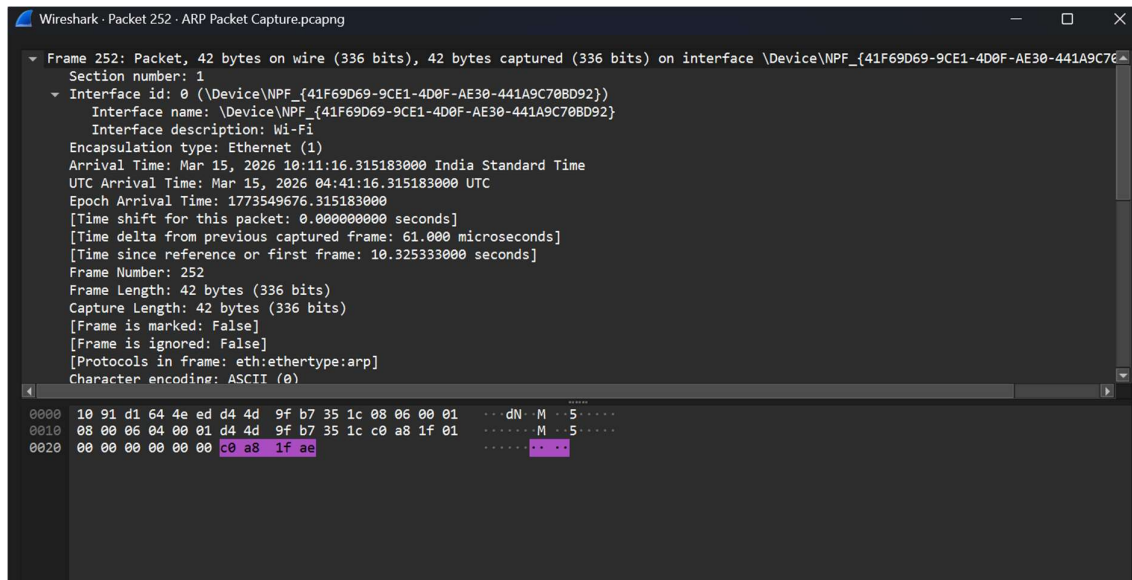
Profile: Default

Module - 6:

1. Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames when your device attempts to find the router's MAC address. Discuss the importance of ARP in packet forwarding.

ARP Packets Capture:

```
▼ Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: SkyworthDigi_b7:35:1c (d4:4d:9f:b7:35:1c)
  Sender IP address: 192.168.31.1
  Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
  Target IP address: 192.168.31.174
```



```
▼ Ethernet II, Src: SkyworthDigi_b7:35:1c (d4:4d:9f:b7:35:1c), Dst: Intel_64:4e:ed (10:91:d1:64:4e:ed)
  Destination: Intel_64:4e:ed (10:91:d1:64:4e:ed)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Source: SkyworthDigi_b7:35:1c (d4:4d:9f:b7:35:1c)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 0]
```

2. Manually configure static routes on a router to direct packets to different subnets. Use the ip route command and verify connectivity using ping and traceroute.

Router Config:

R1:

```
R1
*Mar 17 20:05:32.279: %SYS-5-CONFIG_I: Configured from console by console
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#int f0/0
R1(config-if)#
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#exit
R1(config)#e
*Mar 17 20:06:06.047: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
R1(config)#exit
R1#
*Mar 17 20:06:07.047: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
*Mar 17 20:06:08.043: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#int f1/0
R1(config-if)#
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
*Mar 17 20:07:06.187: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 17 20:07:07.187: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1#
R1#
*Mar 17 20:07:07.639: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#int f1/0
R1(config-if)#ip route 192.168.2.0 255.255.255.2 10.0.0.2
%Inconsistent address and mask
R1(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)#
R1(config)#exit
R1#exit
*Mar 17 20:18:37.059: %SYS-5-CONFIG_I: Configured from console by console
R1#exit
```

R2:

```
R2#en
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#exit
R2(config)#exit
R2#
R2#
*Mar 17 20:21:28.687: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
R2#en
*Mar 17 20:21:29.671: %SYS-5-CONFIG_I: Configured from console by console
*Mar 17 20:21:29.687: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2#en
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#ip address 10.0.0.2 255.255.255.252
R2(config)#
% Invalid input detected at '^' marker.

R2(config)#int f1/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#
R2(config-if)#no shut
R2(config-if)#
R2(config-if)#exit
R2(config)#exit
R2#
R2#sh i
*Mar 17 20:22:36.043: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 17 20:22:37.043: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R2(config)#exit
R2#
R2#sh ip int brief
```

```
R2#sh ip int brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.168.2.1 YES manual up up
FastEthernet1/0 10.0.0.2 YES manual up up
FastEthernet2/0 unassigned YES unset administratively down down
FastEthernet2/1 unassigned YES unset administratively down down
Serial3/0 unassigned YES unset administratively down down
Serial3/1 unassigned YES unset administratively down down
Serial3/2 unassigned YES unset administratively down down
Serial3/3 unassigned YES unset administratively down down
Serial3/4 unassigned YES unset administratively down down
Serial3/5 unassigned YES unset administratively down down
Serial3/6 unassigned YES unset administratively down down
Serial3/7 unassigned YES unset administratively down down
Ethernet4/0 unassigned YES unset administratively down down
Ethernet4/1 unassigned YES unset administratively down down
Ethernet4/2 unassigned YES unset administratively down down
Ethernet4/3 unassigned YES unset administratively down down
Ethernet4/4 unassigned YES unset administratively down down
Ethernet4/5 unassigned YES unset administratively down down
Ethernet4/6 unassigned YES unset administratively down down
Ethernet4/7 unassigned YES unset administratively down down
Serial5/0 unassigned YES unset administratively down down
Serial5/1 unassigned YES unset administratively down down
Serial5/2 unassigned YES unset administratively down down
Serial5/3 unassigned YES unset administratively down down
POS6/0 unassigned YES unset administratively down down
R2#
R2#
R2#
R2#en
R2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f1/0
R2(config-if)#
R2(config-if)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
R2(config)#
```

PC Configs:

```

Welcome to Virtual PC Simulator, version 0.0.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpc.ssf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> show ip

NAME       : PC1[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20010
RHOST:PORT  : 127.0.0.1:20011
MTU         : 1500

```

```

ase visit wiki.freecode.c
om.cn.

Press '?' to get help.

Executing the startup fil
e

PC2> ip 192.168.1.20/24 192.168.1.1
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0 gateway 192.168.1.1

PC2> show ip

NAME       : PC2[1]
IP/MASK     : 192.168.1.20/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 20012
RHOST:PORT  : 127.0.0.1:20013
MTU         : 1500

```

```

om.cn.

Press '?' to get help.

Executing the startup fil
e

PC3>
PC3>
PC3>
PC3> ip 192.168.2.10/24 192.168.2.1
Checking for duplicate address...
PC3 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1

PC3> show ip

NAME       : PC3[1]
IP/MASK     : 192.168.2.10/24
GATEWAY     : 192.168.2.1
DNS         :
MAC         : 00:50:79:66:68:02
LPORT      : 20014
RHOST:PORT  : 127.0.0.1:20015
MTU         : 1500

```

```

Press '?' to get help.

Executing the startup file

PC4> ip 192.168.2.10/24 192.168.2.1
Checking for duplicate address...
192.168.2.10 is being used by MAC 00:50:79:66:68:02
Address not changed

PC4> ip 192.168.2.20/24 192.168.2.1
Checking for duplicate address...
PC4 : 192.168.2.20 255.255.255.0 gateway 192.168.2.1

PC4> show ip

NAME       : PC4[1]
IP/MASK     : 192.168.2.20/24
GATEWAY     : 192.168.2.1
DNS         :
MAC         : 00:50:79:66:68:03
LPORT      : 20016
RHOST:PORT  : 127.0.0.1:20017
MTU         : 1500

```

Ping Test:

Ping from PC1 from Network 1 to PCs from Other Network:

```

MTU       : 1500

PC1> ping 192.168.2.10

84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=366.250 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=258.286 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=355.425 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=219.034 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=336.805 ms

PC1> ping 192.168.2.20

84 bytes from 192.168.2.20 icmp_seq=1 ttl=62 time=467.764 ms
84 bytes from 192.168.2.20 icmp_seq=2 ttl=62 time=267.046 ms
84 bytes from 192.168.2.20 icmp_seq=3 ttl=62 time=273.848 ms
84 bytes from 192.168.2.20 icmp_seq=4 ttl=62 time=187.112 ms
84 bytes from 192.168.2.20 icmp_seq=5 ttl=62 time=299.350 ms

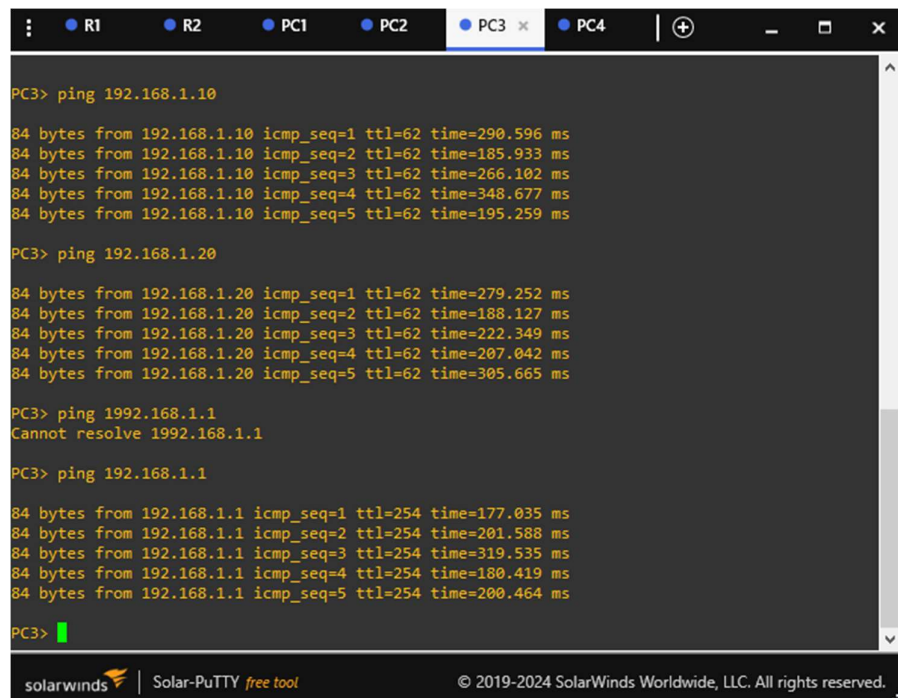
PC1> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=254 time=275.857 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=254 time=151.767 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=254 time=213.930 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=254 time=238.574 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=254 time=156.326 ms

PC1>

```

Ping from PC3 from Network 2 to Other Network PCs:



The screenshot shows a SolarWinds Solar-PuTTY terminal window with tabs for R1, R2, PC1, PC2, PC3 (active), and PC4. The terminal displays the following commands and results:

```
PC3> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=290.596 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=185.933 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=266.102 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=348.677 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=195.259 ms

PC3> ping 192.168.1.20
84 bytes from 192.168.1.20 icmp_seq=1 ttl=62 time=279.252 ms
84 bytes from 192.168.1.20 icmp_seq=2 ttl=62 time=188.127 ms
84 bytes from 192.168.1.20 icmp_seq=3 ttl=62 time=222.349 ms
84 bytes from 192.168.1.20 icmp_seq=4 ttl=62 time=207.042 ms
84 bytes from 192.168.1.20 icmp_seq=5 ttl=62 time=305.665 ms

PC3> ping 1992.168.1.1
Cannot resolve 1992.168.1.1

PC3> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=254 time=177.035 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=254 time=201.588 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=254 time=319.535 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=254 time=180.419 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=254 time=200.464 ms

PC3> 
```

The bottom of the window shows the SolarWinds logo, "Solar-PuTTY free tool", and the copyright notice "© 2019-2024 SolarWinds Worldwide, LLC. All rights reserved."

3. Given a network address of 10.0.0.0/24, divide it into 4 equal subnets.

Calculate the new subnet mask.

Determine the valid host range for each subnet.

Assign IP addresses to devices in Packet Tracer and verify connectivity.

10.0.0.0/24
Considering it to be classless Addressing.
/24

→ 11111111.11111111.11111111.00000000
Net = 255.255.255.0
Mask

∴ Network ID = 24Bit

① 10.0.0.0 0 0 0 0 0 0 0 0
② 10.0.0.0 0 1 - - - - -
③ 10.0.0.0 1 0 - - - - -
④ 10.0.0.0 1 1 - - - - -

Subnet - 1:
Start IP: 10.0.0.0 (Network ID)
End IP: 10.0.0.63 (Broadcast IP)
Subnet mask: /26
255.255.255.252

Subnet - 2:
Start IP: 10.0.0.64 (Network IP)
Broadcast IP: 10.0.0.127 (End IP)
Subnet mask: 255.255.255.252

Subnet - 3:
Start: 10.0.0.128
End: 10.0.0.191
Subnet mask: 255.255.255.252

Subnet-4:

Start: 10.0.0.192

End: 10.0.0.255

Subnetmask: 255.255.255.252

Considering it to be a class A address

∴ Netmask: 255.0.0.0

∴ ① 10. 0 0 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0

② 10. 0 1 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0

③ 10. 1 0 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0

④ 10. 1 1 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0 • 0 0 0 0 0 0 0 0

I: No. of hosts per Subnet: 2^{22} ~~8! × 8! × 6!~~

10. 0.0.0 to 10. 63.255.255

Subnet mask: /10

⇒ 255.192.0.0 ($2^{22} - 2$) No. of hosts.

II:

10. 64.0.0 to 10. 127.255.255

Subnet mask: /10 ⇒ 255.192.0.0

III:

10. 128.0.0 to 10. 191.255.255

Subnet mask: 255.192.0.0

IV:

10. 192.0.0 to 10. 255.255.255

4. You are given three IP addresses: 192.168.10.5, 172.20.15.1, and 8.8.8.8. Identify the class of each IP address. Determine if it is private or public. Explain how NAT would handle a private IP when accessing the internet.

i) 192.168.10.5 – Class C

Start IP: 192.168.10.0 (Network ID)

End IP: 192.168.10.255 (Broadcast Address)

Usable Host Addresses: 192.168.10.1 to 192.168.10.254

Subnet Mask: 255.255.255.0

ii) 172.20.15.1 – Class B

Start IP: 172.20.0.0 (Network ID)

End IP: 172.20.255.255 (Broadcast Address)

Usable Host Addresses: 172.20.0.1 to 172.20.255.254

Subnet Mask: 255.255.0.0

iii) 8.8.8.8 – Class A

Start IP: 8.0.0.0 (Network ID)

End IP: 8.255.255.255 (Broadcast Address)

Usable Host Addresses: 8.0.0.1 to 8.255.255.254

Subnet Mask: 255.0.0.0

5. In Cisco Packet Tracer, configure NAT on a router to allow internal devices (192.168.1.x) to access the internet. Test connectivity by pinging an external public IP.

Capture the traffic in Wireshark and analyze the source IP before and after NAT translation.

Router Config:

```
R1#
R1#
R1#
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
*Mar 18 19:01:10.351: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 18 19:01:11.351: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#exit
R1#
*Mar 18 19:01:27.539: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
*Mar 18 19:02:07.495: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 18 19:02:08.495: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1#
R1#
*Mar 18 19:02:08.803: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          192.168.1.1     YES manual up          up
FastEthernet0/1          192.168.2.1     YES manual up          up
FastEthernet1/0          unassigned      YES unset  administratively down down
R1#
```

Configure the NAT Interface of the Router with DHCP Protocol:

```
R1
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f1/0
R1(config-if)#
R1(config-if)#ip address dhcp
R1(config-if)#no shut
R1(config-if)#
*Mar 18 19:10:45.283: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to u
p
*Mar 18 19:10:46.283: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1
/0, changed state to up
R1(config-if)#
R1(config-if)#
*Mar 18 19:10:57.259: %DHCP-6-ADDRESS_ASSIGN: Interface FastEthernet1/0 assigned DH
CP address 192.168.122.7, mask 255.255.255.0, hostname R1

R1(config-if)#exit
R1(config)#exit
R1#
R1#sh
*Mar 18 19:11:07.331: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int brief
Interface                IP-Address      OK? Method Status      Protoco
l
FastEthernet0/0          192.168.1.1     YES manual up          up
FastEthernet0/1          192.168.2.1     YES manual up          up
FastEthernet1/0          192.168.122.7   YES DHCP  up          up
```

PC Config:

```
192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 20017
RHOST:PORT : 127.0.0.1:20018
MTU        : 1500

PC1>

192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2> ip 192.168.1.20/24 192.168.1.1
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0 gateway 192.168.1.1

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 192.168.1.20/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 20011
RHOST:PORT : 127.0.0.1:20012
MTU        : 1500

PC2>

192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC3> ip 192.168.2.10/24 192.168.2.1
Checking for duplicate address...
PC3 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1

PC3> show ip

NAME       : PC3[1]
IP/MASK    : 192.168.2.10/24
GATEWAY    : 192.168.2.1
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 20013
RHOST:PORT : 127.0.0.1:20014
MTU        : 1500

PC3>

192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC4> ip 192.168.2.20/24 192.168.2.1
Checking for duplicate address...
PC4 : 192.168.2.20 255.255.255.0 gateway 192.168.2.1

PC4> show ip

NAME       : PC4[1]
IP/MASK    : 192.168.2.20/24
GATEWAY    : 192.168.2.1
DNS        :
MAC        : 00:50:79:66:68:03
LPORT     : 20015
RHOST:PORT : 127.0.0.1:20016
MTU        : 1500

PC4>
```

Configure NAT Inside Local Networks:

```
R1#
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#
R1(config-if)#ip nat inside
R1(config-if)#
*Mar 18 19:12:53.031: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, changed state to up
R1(config-if)#exit
R1(config)#
R1(config)#exit
R1#
R1#en
*Mar 18 19:13:01.675: %SYS-5-CONFIG_I: Configured from console by console
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/1
R1(config-if)#ip nat inside
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
R1#
*Mar 18 19:13:20.703: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
```

Configure Outside NAT:

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#int f1/0
R1(config-if)#ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#
R1#
*Mar 18 19:14:24.627: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Create Access List:

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)#access-list 1 permit 192.168.2.0 0.0.0.255
R1(config)#
```

Perform NAT Overload:

```
R1(config)#ip nat inside source list 1 interface FastEthernet1/0 overload
R1(config)#ip route 0.0.0.0 0.0.0.0 FastEthernet1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R1(config)#exir
```

Add Default Gateway:

```
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.122.1
R1(config)#
R1(config)#exit
R1#
R1#
*Mar 18 19:25:59.975: %SYS-5-CONFIG_I: Configured from console by console
R1#ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/91/224 ms
R1#
```

Ping Test to a Global IP from All PCs:

The image displays four terminal windows, each representing a different PC (PC1, PC2, PC3, and PC4) connected to a network at 192.168.11.128. Each window shows the results of a ping test to the global IP address 8.8.8.8. The windows are arranged in a 2x2 grid.

PC1 Terminal: Shows five successful ping attempts to 8.8.8.8. The first three attempts are successful with varying round-trip times (e.g., 78.272 ms, 58.453 ms, 124.693 ms). The last two attempts are successful with round-trip times of 53.502 ms and 61.976 ms.

PC2 Terminal: Shows five successful ping attempts to 8.8.8.8. The first three attempts are successful with varying round-trip times (e.g., 98.367 ms, 83.560 ms, 86.105 ms). The last two attempts are successful with round-trip times of 72.362 ms and 75.899 ms.

PC3 Terminal: Shows five successful ping attempts to 8.8.8.8. The first three attempts are successful with varying round-trip times (e.g., 220.165 ms, 76.976 ms, 76.976 ms). The last two attempts are successful with round-trip times of 76.976 ms and 76.976 ms.

PC4 Terminal: Shows five successful ping attempts to 8.8.8.8. The first three attempts are successful with varying round-trip times (e.g., 54.128 ms, 68.019 ms, 68.019 ms). The last two attempts are successful with round-trip times of 68.019 ms and 68.019 ms.

Packet Capture From Wireshark:

*Standard Input [R1 FastEthernet1/0 to NAT1 nat0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
30	20.011079	8.8.8.8	192.168.122.7	ICMP	98	Echo (ping) reply id=0xeeaf, seq=3/768, ttl=127 (request in 28)
31	21.196546	192.168.122.7	8.8.8.8	ICMP	98	Echo (ping) request id=0xefaf, seq=2/512, ttl=63 (reply in 32)
32	21.246697	8.8.8.8	192.168.122.7	ICMP	98	Echo (ping) reply id=0xefaf, seq=2/512, ttl=127 (request in 31)
33	21.988093	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
34	22.083796	ca:01:6b:00:00:1c	ca:01:6b:00:00:1c	LOOP	60	Reply
35	23.030869	192.168.122.7	8.8.8.8	ICMP	98	Echo (ping) request id=0xf1af, seq=5/1280, ttl=63 (reply in 36)
36	23.066646	8.8.8.8	192.168.122.7	ICMP	98	Echo (ping) reply id=0xf1af, seq=5/1280, ttl=127 (request in 35)
37	23.974615	ca:01:6b:00:00:1c	Broadcast	ARP	60	Who has 8.8.8.8? Tell 192.168.122.7
38	24.000050	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
39	25.974144	192.168.122.7	8.8.8.8	ICMP	98	Echo (ping) request id=0xf4af, seq=3/768, ttl=63 (reply in 41)
40	25.988197	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
41	26.034715	8.8.8.8	192.168.122.7	ICMP	98	Echo (ping) reply id=0xf4af, seq=3/768, ttl=127 (request in 39)
42	28.090169	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
43	29.070818	192.168.122.7	8.8.8.8	ICMP	98	Echo (ping) request id=0xf7af, seq=5/1280, ttl=63 (reply in 44)
44	29.112397	8.8.8.8	192.168.122.7	ICMP	98	Echo (ping) reply id=0xf7af, seq=5/1280, ttl=127 (request in 43)
45	29.984837	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
46	32.000842	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
47	32.109176	ca:01:6b:00:00:1c	ca:01:6b:00:00:1c	LOOP	60	Reply
48	33.055600	ca:01:6b:00:00:1c	CDP/VTP/DTP/PagP/UD...	CDP	382	Device ID: R1 Port ID: FastEthernet1/0
49	33.988089	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
50	36.000199	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
51	37.084203	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
52	40.000952	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
53	41.983948	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001
54	42.096830	ca:01:6b:00:00:1c	ca:01:6b:00:00:1c	LOOP	60	Reply
55	44.000413	26:23:ff:09:14:46	Nearest-Customer-Br...	STP	52	Conf. Root = 32768/0/52:54:00:a1:07:64 Cost = 0 Port = 0x8001

Frame 1: Packet, 52 bytes on wire (416 bits), 52 bytes captured (416 bits) on interface

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

wireshark_Standard InputUHQ9L3.pcapng

Packets: 55 · Dropped: 0 (0.0%)

Profile: Default